KONGU ENGINEERING COLLEGE

PERUNDURAI ERODE-638060

DEPARTMENT OF CT-PG

Transform Yourself

TEXMART - SMART TEXTILE MANAGEMENT SYSTEM

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SRI POONGUZHALI COTTON MILLS

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Objectives

- To automate mill operations and reduce manual work.
- To track inventory, products, and suppliers in real time.
- To minimize errors and data duplication.
- To provide a user-friendly centralized system.
- To improve overall efficiency and decision-making in the mill.

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Existing System

- Operations like product handling and stock are managed manually.
- Records on paper cause errors and duplication.
- No real-time tracking of stock or product details.
- Fabric quality and licenses are handled inefficiently.
- Manual processes slow down overall operations.

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Proposed System

- Centralized digital system for all mill operations.
- Automation of production, inventory, and billing tasks.
- Real-time data access with improved accuracy.
- Easy-to-use interface for staff.
- Secure access with authentication and roles.

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Modules

The modules used in our project are:

- Admin Panel Module
- Supplier and Purchase Module
- Stock Management Module
- Product Display Module
- About Us (License, Fabric Quality) Module

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Sub-Modules:

Admin Panel Module:

- Admin login and authentication
- User and role management
- Product add / update / delete
- Dashboard and reports management

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Sub-Modules:

Supplier and Purchase Module:

- Supplier registration and management
- Purchase order creation
- Purchase history tracking
- Invoice and payment records

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Sub-Modules:

Stock Management Module:

- Stock entry and updates
- Real-time stock level monitoring
- Low stock alerts
- Stock reports and history

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Sub-Modules:

Product Display Module:

- Product listing and categorization
- Product details view (fabric type, quality, price)
- Search and filter products
- Product availability status

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Sub-Modules:

About Us (License, Fabric Quality) Module:

- Company profile management
- License and certification details
- Fabric quality standards information
- Compliance and policy display

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Technologies Used

Front End:

- HTML
- Tailwind CSS
- React. js

Back End:

- Node.js
- Express.js

Database:

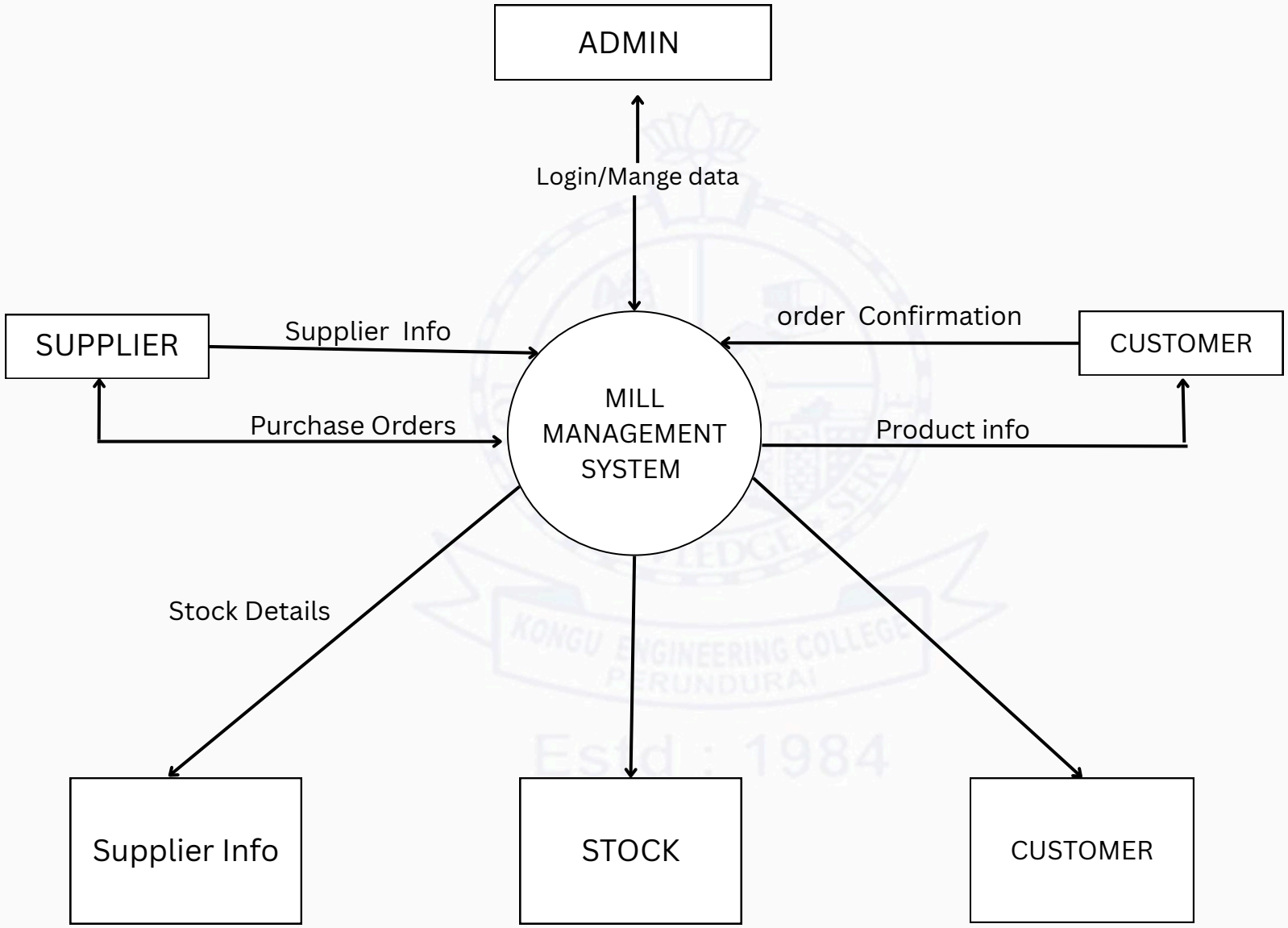
- Mongo DB

Payment Gateway:

- RazorPay

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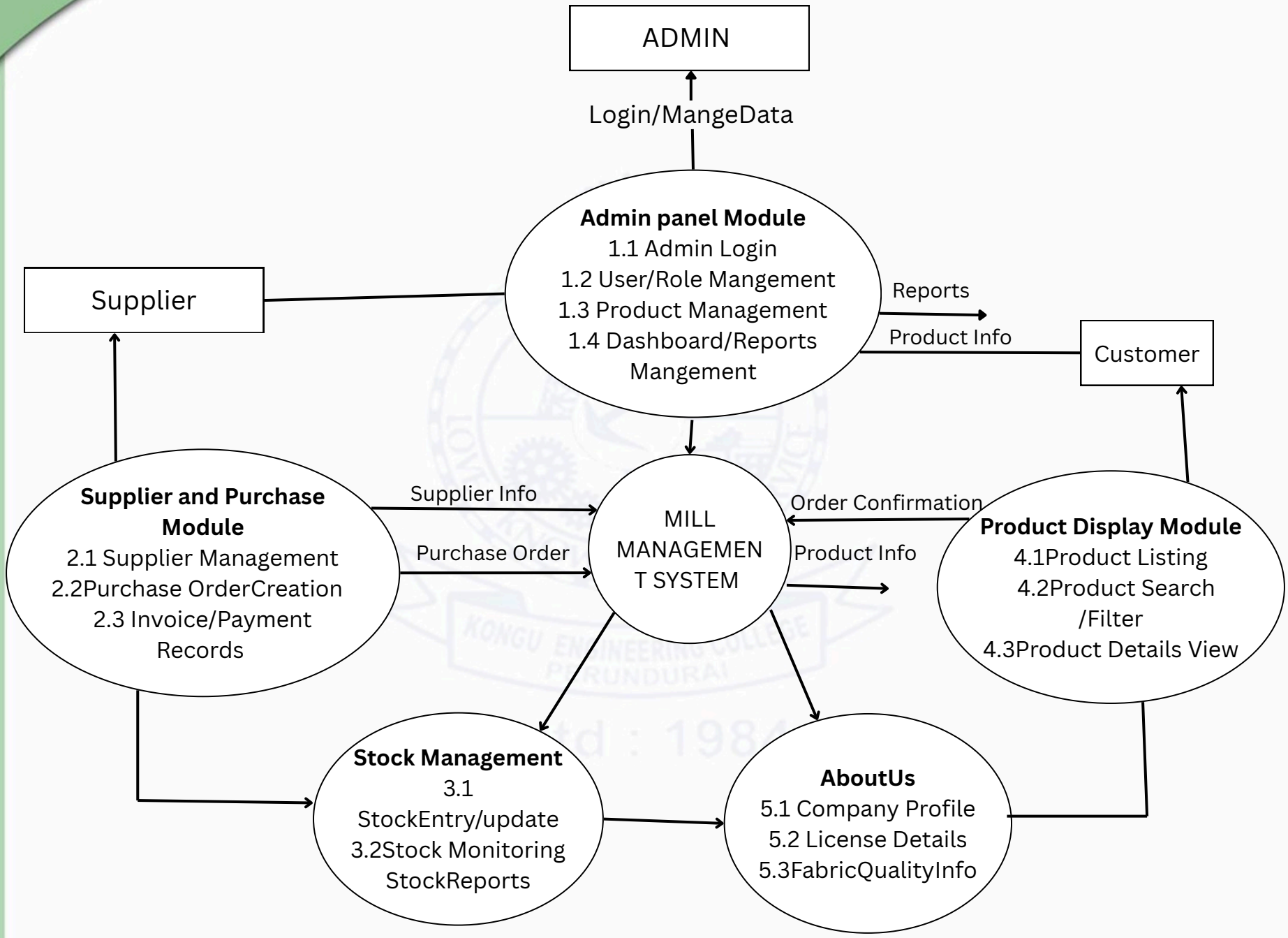
Data flow diagram for level 0



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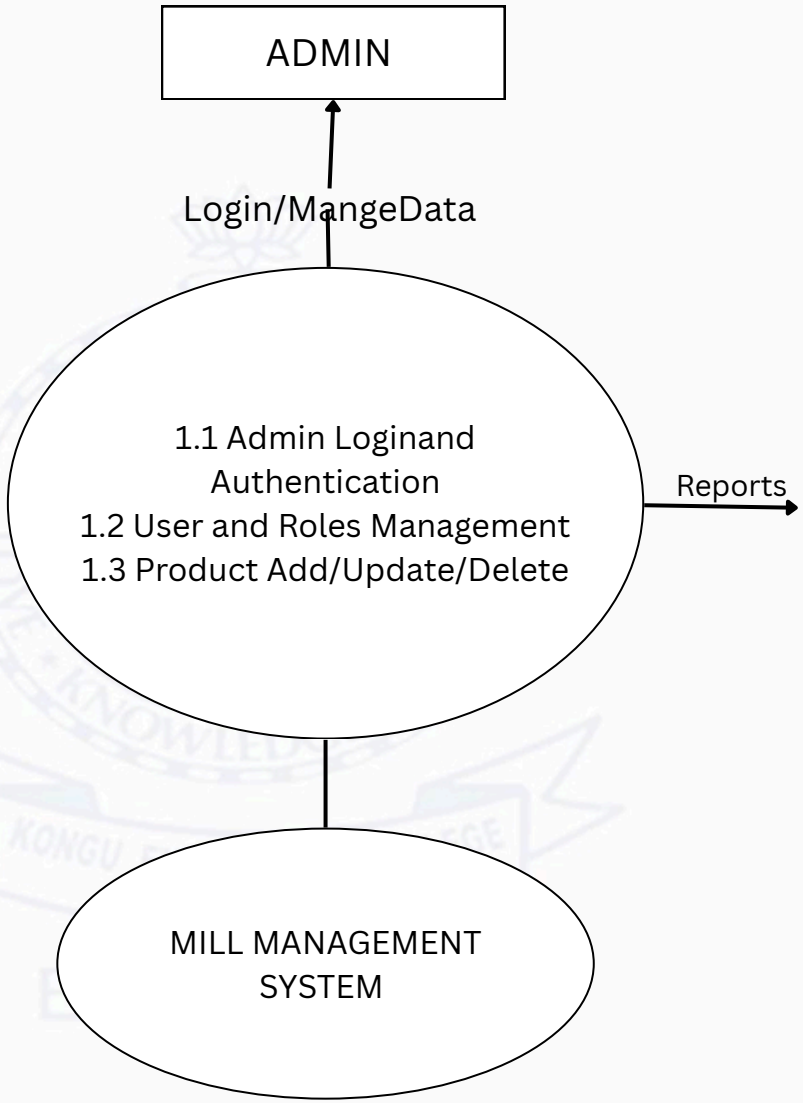
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Data flow diagram for level 1

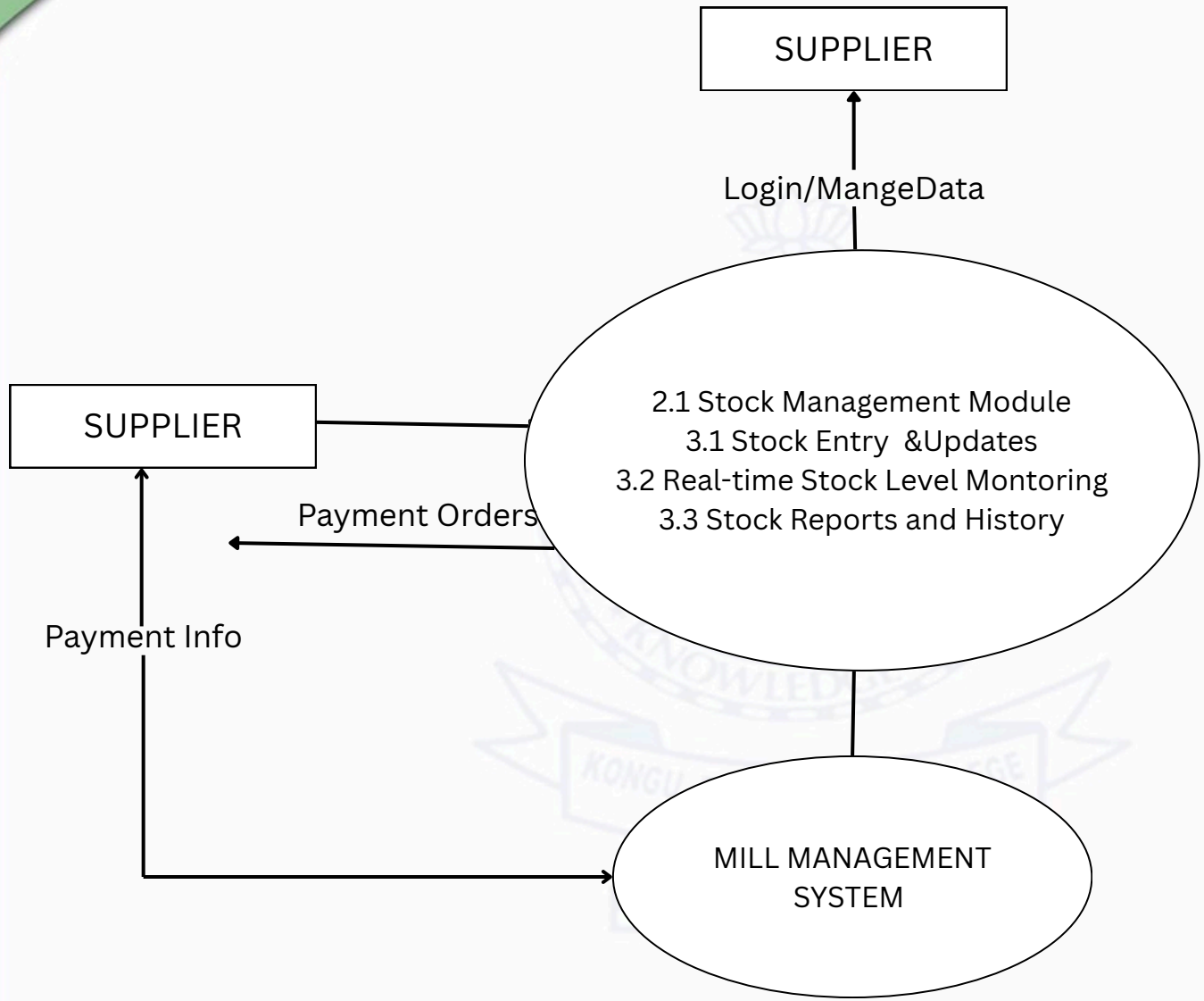


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DFD Level 2(Admin Panel Module)

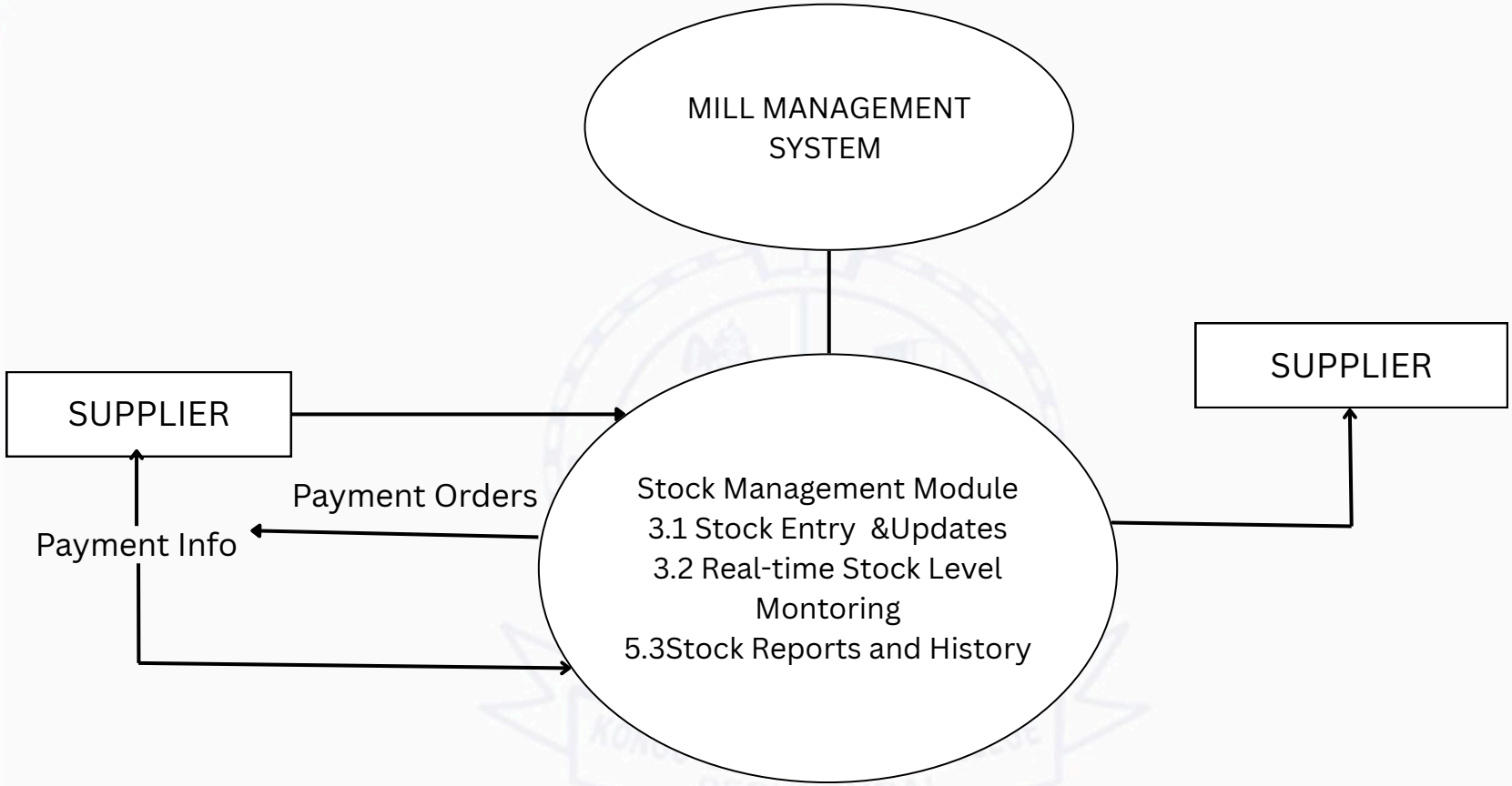


DFD Level 2(Supplier anad Purchase Module)



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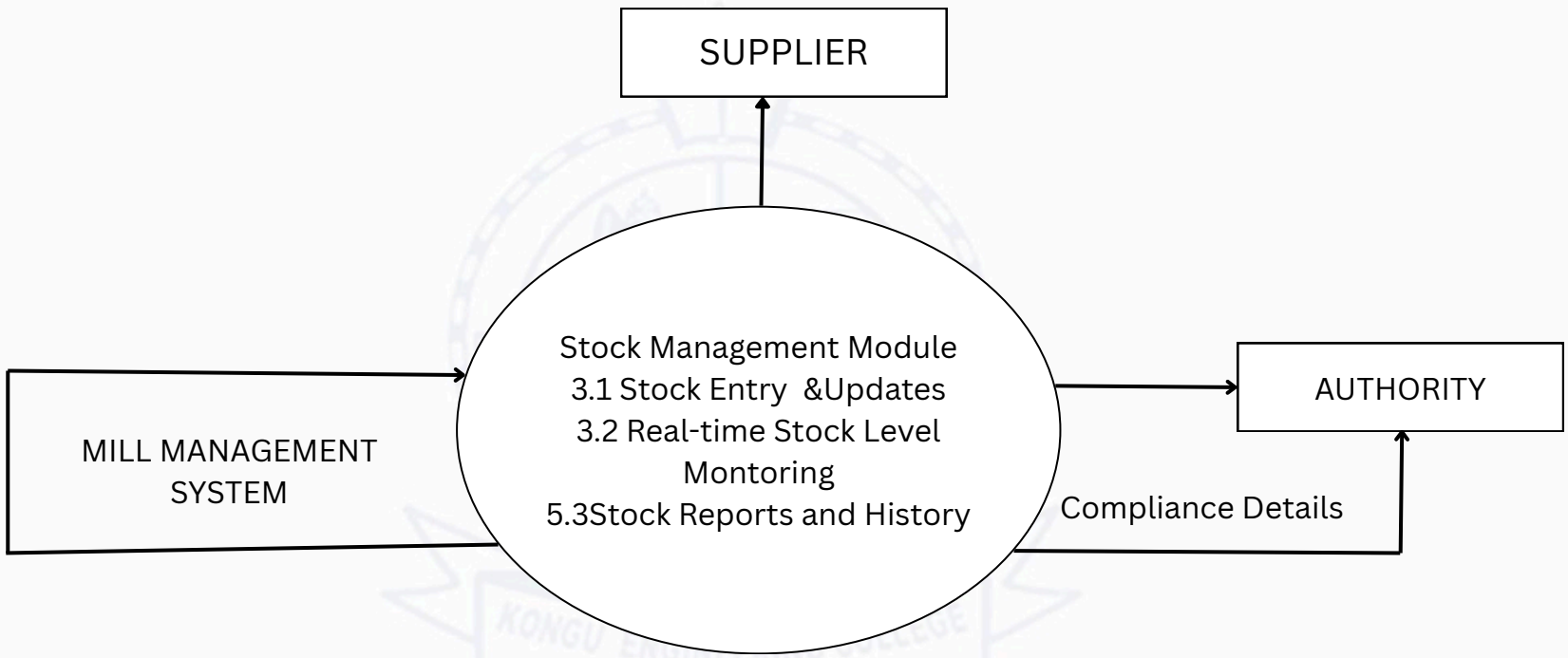
DFD Level 2(Stock Management Module)



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DFD Level 2(Product Display Module)



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System Features Details

TABLES

- System Features Details
- Modules Used Details
- Admin Panel Module Details
- Supplier & Purchase Module Details
- Stock Management Module Details
- Product Display Module Details
- About Us (License & Fabric Quality) Module Details

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System Features Details

FIELD NAME	DATA TYPE	SIZE	CONSTRAINTS
feature_id	INTEGER	5	PRIMARY KEY
system_type	VARCHAR	50	NOT NULL
automation_tasks	VARCHAR	150	NOT NULL
real_time_access	VARCHAR	50	NOT NULL
user_interface	VARCHAR	50	NOT NULL
security_access	VARCHAR	50	NOT NULL

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Modules Used Details

FIELD NAME	DATA TYPE	SIZE	CONSTRAINTS
module_id	INTEGER	5	PRIMARY KEY
admin_panel_module	VARCHAR	50	NOT NULL
supplier_purchase_module	VARCHAR	50	NOT NULL
stock_management_module	VARCHAR	50	NOT NULL
product_display_module	VARCHAR	50	NOT NULL
about_us_module	VARCHAR	80	NOT NULL

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Admin Panel Module Details

FIELD NAME	DATA TYPE	SIZE	CONSTRAINTS
admin_id	**DATA TYPE**	5	PRIMARY KEY
admin_login	**DATA TYPE**	50	NOT NULL
user_role_management	**DATA TYPE**	100	NOT NULL
product_management	**DATA TYPE**	100	NOT NULL
dashboard_reports	**DATA TYPE**	100	NOT NULL

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Supplier & Purchase Module Details

FIELD NAME	DATA TYPE	SIZE	CONSTRAINTS
supplier_id	INTEGER	5	PRIMARY KEY
supplier_registration	VARCHAR	100	NOT NULL
purchase_order	VARCHAR	100	NOT NULL
purchase_history	VARCHAR	150	NOT NULL
invoice_payment	VARCHAR	100	NOT NULL

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Stock Management Module Details

FIELD NAME	DATA TYPE	SIZE	CONSTRAINTS
stock_id	INTEGER	5	PRIMARY KEY
stock_entry	VARCHAR	100	NOT NULL
stock_monitoring	VARCHAR	100	NOT NULL
low_stock_alert	VARCHAR	50	NOT NULL
stock_reports	VARCHAR	100	NOT NULL

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Product Display Module Details

FIELD NAME	DATA TYPE	SIZE	CONSTRAINTS
product_id	INTEGER	5	PRIMARY KEY
product_listing	VARCHAR	100	NOT NULL
product_details	VARCHAR	150	NOT NULL
search_filter	VARCHAR	100	NOT NULL
availability_status	VARCHAR	50	NOT NULL

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About Us (License & Fabric Quality) Module Details

FIELD NAME	DATA TYPE	SIZE	CONSTRAINTS
about_id	INTEGER	5	PRIMARY KEY
company_profile	VARCHAR	150	NOT NULL
license_details	VARCHAR	100	NOT NULL
fabric_quality	VARCHAR	100	NOT NULL
compliance_policy	VARCHAR	100	NOT NULL

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Future Enhancements:

- Add cloud backup to store data safely and access it anytime
- Develop a mobile app to check system updates instantly
- Apply smart AI tools to support better planning and decisions
- Introduce a billing system that works automatically
- Use barcode or RFID to manage stock quickly and accurately
- Strengthen security with fingerprint or face login
- Upgrade the system to be smoother, safer, and easier to use

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Conclusion:

- The system helps manage all mill work in one place
- It reduces manual work by automating tasks
- It improves accuracy and saves time
- Real-time data makes work faster and easier
- It increases productivity and efficiency
- The system makes mill management modern and reliable

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Thank You

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